BIDYA BHUSHAN NANDA

Full-Stack & BSP Engineer | Mobile, Systems, AI Integration

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SUMMARY

Engineer who works from bare silicon up to shipped software. Full-stack apprentice at AB Connectz building HOSHCARE (in development), founder of Broccoli, and B.Tech Electronics student (MAIT, 2029).

PROOF AT A GLANCE

Broccoli	Live on Google Play: about 100 users, 5-star rating from 20+ reviews, growing quickly	live on Google Play | case study
HOSHCARE	~19-screen health-records app; AI document extraction tested on 500+ varied documents	In development; closed-source, walkthrough on request
Zucchini Linux	Arch Linux ARM and LineageOS dual-boot on a locked-down Qualcomm tablet	Research paper | project
Kestrel	Rust ground control station for PX4 and ArduPilot, tested on SITL	Case study
Habanero	Fully offline multi-agent engineering engine on Apple Silicon	Case study

EXPERIENCE

AB Connectz — Remote Jun 2026 – Present

Full-Stack Engineer, Paid Apprenticeship (Intern, Jun – Jul 2026) Jul 2026 – Present

HOSHCARE: health-records app I develop (in development; closed-source; walkthrough on request).

- Own the React Native / Expo (SDK 56) app (~19 screens); review-before-save and permanent record locking ensure extracted data is never written without user confirmation.
- Built an AI pipeline that turns handwritten medical documents into structured JSON (schema-constrained output, exponential-backoff retries, uncertainty flag for illegible text), tested on 500+ varied documents from lab reports and prescriptions to false data. Hard part: fixing an underspecified schema that made the model return empty objects.
- Built the original REST API (Node.js, Express, Prisma, PostgreSQL), then moved the client onto the shared NestJS backend (JWT/OTP auth) and removed Firebase Auth; built the web app on Supabase with row-level security, verified so no user can reach another's records or storage.
- Shipped the 8-page marketing site (Three.js, WebGL, Lenis), whose design system carried into the product app; fixed a silent auth failure (breaking library update) and an upload failure (Expo's fetch polyfill mishandling multipart data).

Broccoli — Founder & Developer, independent venture Aug 2026 – Present

- Founded a nutrition-tracking startup, newly in production on Google Play (details under Projects).

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Rust, C/C++, SQL, GLSL
Mobile & Frontend: React Native, Expo, React, Next.js, Svelte, Tauri, Tailwind CSS, Three.js, WebGL
Backend & Data: Node.js, NestJS, Express, Prisma, Drizzle, REST APIs, JWT auth, PostgreSQL, SQLite, Supabase (RLS)
AI & ML: Gemini API, LLM document extraction, schema-constrained prompting, multi-agent orchestration, local LLMs (GGUF)
BSP & Linux: ARM64 kernel bring-up, bootloaders (S-Boot, LK2ND), device tree, initramfs, vbmeta / dm-verity, Arch Linux ARM, LineageOS
UAV & Embedded: MAVLink v2, PX4, ArduPilot, SITL, Arduino, ESP32, Raspberry Pi, camera and sensor interfacing
Tools: Git, GitHub, Docker, EAS Build, Playwright, fastboot, Odin

EDUCATION

B.Tech, Electronics Engineering — MAIT, Delhi 2025 – 2029 (expected)

CGPA 8.5 / 10 | Second year. Software and systems skills self-taught from official documentation and applied in product work.

ALSO

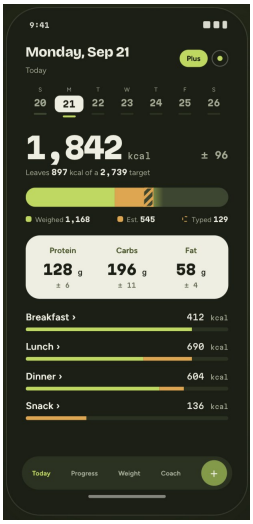
- Portfolio site bidyabhushan.in (Next.js 16, Three.js) and the paper "Augmentation, Not Obsolescence" on AI and entry-level developer jobs.
- Aerostars UAV Society, MAIT — member, drones, IoT and embedded systems (Sep 2025 – Present). Photography and film as unfocused.nerd.

PROJECTS

Broccoli — Offline-First Nutrition Tracker (live on Google Play) Aug 2026 – Present

React Native (Expo SDK 57), TypeScript, SQLite, Drizzle, NestJS, PostgreSQL | bidyabhushan.in/broccoli

- Shows totals as honest ranges, not false-precision numbers; works offline on SQLite, with South Asian food coverage.
- Built AI food recognition, barcode and label capture, 60 FPS dial interactions (Reanimated) and Skia charts with an adaptive TDEE model.
- Hard part: security-audited my own auth flow, found a deep-link password-reset token hijack risk and specified the verified App Links fix.
- Result: live on Google Play with about 100 users and a 5-star rating from 20+ reviews, growing quickly.

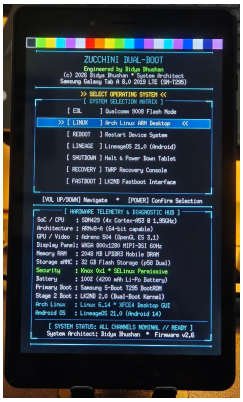


Today screen, with totals as ranges

Zucchini Linux — Bare-Metal ARM64 Bring-Up (Samsung SM-T295) Oct 2026 – Present

C, Linux kernel, LK2ND, S-Boot, device tree, Arch Linux ARM, LineageOS | bidyabhushan.in/projects/zucchini-linux

- Dual-booted native Arch Linux ARM and LineageOS on a locked-down Qualcomm SDM429 tablet by reverse-engineering the S-Boot chain and adding the LK2ND bootloader.
- Hard part: a warm-reboot “SECURE CHECK FAIL: vbmeta” bootloop, fixed with a verification-disabled vbmeta image after SHA-256-verified off-device backups of efs, persist and modem.
- Rebuilt the kernel with quota and dm-verity support after the Arch rootfs failed to mount in initramfs.
- Validated the LineageOS kernel and storage device tree with RAM-boot guard images that keep eMMC read-only, so the Arch install was never at risk.
- Research paper: bidyabhushan.in/blog/sm-t295-linux-bringup



Zucchini boot menu running on the SM-T295

Kestrel — Custom Drone Ground Control Station Oct 2026 – Present

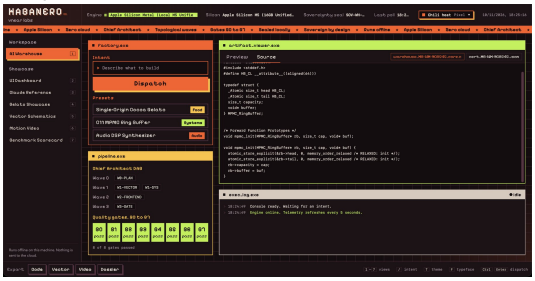
Rust, Tokio, Tauri v2, Svelte 5, MAVLink v2, PX4, ArduPilot, MediaMTX, PMTiles | bidyabhushan.in/projects/kestrel

- Isolates flight-critical MAVLink ingestion (CRC validation, heartbeat supervision) in a Rust/Tokio core, apart from the UI, and normalizes PX4 and ArduPilot flight modes.
- Hard part: safety in the backend. Arming is blocked on stale or failing GPS, EKF, battery or RC-link data, and commands stay pending until acknowledged (2.5 s timeout, never replayed after a reconnect).
- Built for 50 Hz telemetry with a lock-free state cache and binary IPC; video and offline maps run as isolated sidecars; tested on PX4 and ArduPilot SITL.

Habanero — Local Multi-Agent AI Engineering Engine Nov 2026 – Present

TypeScript, Python, C11, Playwright, SQLite FTS5, quantized local LLMs (GGUF) | bidyabhushan.in/projects/habanero

- Fully offline on Apple Silicon: a planner agent splits tasks into work orders run in dependency order on local LLMs, with SHA-256 contract freezing between agents.
- 8-gate verification (Clang C11 builds, contrast and layout checks, Playwright audits) with a capped self-repair loop; passes 25/25 tasks on my own benchmark suite.
- SQLite FTS5 retrieval over verified code patterns, a CLI (run, factory, serve) and a live web studio.



Habanero studio: pipeline with 8 of 8 gates passing

IoT QR-Code Scanner

Arduino, ESP32, Raspberry Pi 3B, C++, Python

- Built a QR-code scanner on Arduino and ESP32 (C++) and Raspberry Pi 3B (Python), covering camera capture, decoding and hardware I/O.